2.3 REPRESENTATION OF SIGNED NUMBERS AND BINARY ARITHMETIC IN COMPUTERS

So far, we have considered only positive numbers. The representation of negative numbers is also equally important. There are two ways of representing signed numbers—sign-magnitude form and complement form. There are two complement forms: 1's complement form and 2's complement form.

Most digital computers do subtraction by the 2's complement method, but some do it by he 1's complement method. The advantage of performing subtraction by the complement method is reduction in the hardware. Instead of having separate digital circuits for addition and subtraction, only adding circuits are needed. That is, subtraction is also performed by adders only. Instead of subtracting one number from the other, the complement of the subtrahend is added to the minuend.

In sign-magnitude form, an additional bit called the *sign bit* is placed in front of the number. If the sign bit is a 0, the number is positive. If it is a 1, the number is negative. For example,

0	1	0	1	0	0	1	= + 41
Sign bit	Magnitude						

1	1	0	1	0	0	1	= – 41
Sign bit	Magnitude						

Under the sign-magnitude system, a great deal of manipulation is necessary to add a positive number to a negative number. Thus, though the sign-magnitude number system is possible, it is impractical.

2.3.1 Representation of Signed Numbers Using the 2's (or 1's) Complement Method

The 2's (or 1's) complement system for representing signed numbers works like this:

1. If the number is positive, the magnitude is represented in its true binary form and a sign bit 0 is placed in front of the MSB.
2. If the number is negative, the magnitude is represented in its 2's (or 1's) complement form and a sign bit 1 is placed in front of the MSB.

That is, to represent the numbers in sign 2's (or 1's) complement form, determine the 2's (or 1's) complement of the magnitude of the number and then attach the sign bit.

The 2's (or 1's) complement operation on a signed number will change a positive number to a negative number and vice versa. The conversion of complement to true binary is the same as the process used to convert true binary to complement. The representation of + 51 and – 51 in both 2's and 1's complement forms is shown below:

0	1	1	0	0	1	1
Sign bit	Magnitude					

= + 51 (In sign magnitude form)
(In sign 2's complement form)
(In sign 1's complement form)

1	1	1	0	0	1	1
Sign bit	Magnitude					

= – 51 (In sign magnitude form)

1	0	0	1	1	0	1
Sign bit	Magnitude					

= – 51 (In sign 2's complement form)

1	0	0	1	1	0	0
Sign bit	Magnitude					

= – 51 (In sign 1's complement form)

Methods of obtaining the 2's complement of a number: The 2's complement of a number can be obtained in three ways as given below.

1. By obtaining the 1's complement of the given number (by changing all 0s to 1s and 1s to 0s) and then adding 1.

modulus arithmetic. For example: 1100 + 1111 = 1011.

In the 2's complement subtraction, add the 2's complement of the subtrahend to the minuend. If there is a carry out, ignore it. Look at the sign bit, i.e. MSB of the sum term. If the MSB is a 0, the result is positive and is in true binary form. If the MSB is a 1 (whether there is a carry or no carry at all) the result is negative and is in its 2's complement form. Take its 2's complement to find its magnitude in binary.

EXAMPLE 2.23 Subtract 14 from 46 using the 8-bit 2's complement arithmetic.

Solution

+ 1 4	=	0 0 0 0 1 1 1 0	
− 1 4	=	1 1 1 1 0 0 1 0	(In 2's complement form)

+ 4 6		0 0 1 0 1 1 1 0	
− 1 4	⇒	+ 1 1 1 1 0 0 1 0	(2's complement form of − 14)
+ 3 2		1 0 0 1 0 0 0 0 0	(Ignore the carry)

There is a carry, ignore it. The MSB is 0. So, the result is positive and is in normal binary form. Therefore, the result is + 00100000 = + 32.

EXAMPLE 2.24 Add − 75 to + 26 using the 8-bit 2's complement arithmetic.

Solution

+ 7 5	=	0 1 0 0 1 0 1 1	
− 7 5	=	1 0 1 1 0 1 0 1	(In 2's complement form)

+ 2 6		0 0 0 1 1 0 1 0	
− 7 5	⇒	+ 1 0 1 1 0 1 0 1	(2's complement form of − 75)
− 4 9		1 1 0 0 1 1 1 1	(No carry)

There is no carry, the MSB is a 1. So, the result is negative and is in 2's complement form. The magnitude is 2's complement of 11001111, that is, 00110001 = 49. Therefore, the result is − 49.